Fair Early Warning in Online Education with Dynamic Bayesian Networks

ICT 5143: Algorithms Fairness and Graphical Models Laboratory
Problem 3; Team 3
Anisha Chandra, 261090830001; Dhanya Padasalgi, 261090830015
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Abstract

We ask whether a temporally regularized alert rule reduces disparities in early support at fixed capacity. On real OULAD data (module BBB, five student-level splits), a three-state dynamic Bayesian network is within 0.03 average precision of logistic regression. The temporal rule raises cumulative detection over per-checkpoint equalisation in every split (+0.12 on test) and lowers the mean coverage gap (0.095 vs 0.114 on test; 0.138 vs 0.253 on transfer). However, the test gap difference is significant on no split, false-alert exposure nearly doubles (0.300 vs 0.166), and the rule does not beat top-k on the gap. The fairness hypothesis is not supported.

1 Question, context and contribution

Stakeholders are students, tutors, disability-support teams, academic advisers and university learning-analytics teams. We set $Y = 1$ for final result Fail or Withdrawn and $Y = 0$ for Pass or Distinction; Fail and Withdrawn are also reported separately. $D_t = 1$ means the student is offered voluntary support at checkpoint $t \in \{2, 4, 6, 8\}$ weeks.

Research question: Can a temporally regularized alert rule reduce disparities in early support while retaining predictive utility and a fixed support budget? Expectation at 20% capacity: the rule lowers the cumulative coverage gap relative to independent per-checkpoint equalisation, and cumulative detection of $Y = 1$ falls by no more than 0.02.

Contribution: A threshold-offset rule with a re-alert discount, tuned jointly across checkpoints on validation data with smoothness penalties on offset changes. It is refuted if the cumulative coverage gap is not lower than the baseline, or if detection falls by more than 0.02 on the held-out or transfer presentation. Detection does not fall in any split, and the mean gap is lower, but the gap reduction is not significant on test, and false-alert exposure exceeds the stated margin (Section 4).

Closest work: Per-checkpoint equal-opportunity threshold adjustment in the style of Hardt et al. [1], reproduced as the `indep_eopp` baseline. Our difference is joint tuning across checkpoints with burden, timing and smoothness terms. Table 1 compares ten closely related papers.

Table 1: Literature review: closest work on fairness, graphical models and early warning.

Reference	Contribution	Method and data	Link to this project
[1] Hardt et al. (2016)	Defines equal opportunity and gives a post-processing rule that equalises true positive rates across groups.	Group-specific thresholds on a fixed score; evaluated on credit and recidivism data.	Closest method; our `indep_eopp` baseline is this rule applied per checkpoint.
[2] Kleinberg et al. (2017)	Shows calibration, balance for positives and balance for negatives cannot all hold together unless base rates are equal.	Theoretical impossibility result for risk scores.	Explains why we report several criteria and why the choice of criterion matters.
[3] Chouldechova (2017)	Shows the same tension for recidivism tools; calibrated scores can produce unequal false positive rates.	Analysis of a risk instrument across groups.	Supports our reporting of false-alert exposure alongside detection.
[4] Kusner et al. (2017)	Proposes counterfactual fairness, based on a causal model.	Causal graph with interventions on protected attributes.	Contrast: we avoid causal claims because our graph assumptions are not justified.
[5] Zafar et al. (2017)	Constrains classifiers to avoid disparate treatment and disparate impact.	Constrained optimisation on logistic models.	Related use of constraints in fairness-aware training; we instead tune post-hoc offsets.
[6] Barocas et al. (2019)	Textbook on fairness definitions, their limits and their trade-offs.	Survey and worked examples.	Background for the definitions in Section 2 and the limits in Section 5.
[7] Murphy (2002)	Thesis on dynamic Bayesian networks: representation, exact inference and EM learning.	Forward filtering and smoothing for hidden Markov and DBN models.	Source of the forward filtering and EM procedure used in our DBN.
[8] Kizilcec et al. (2013)	Identifies distinct patterns of disengagement among MOOC learners.	Clustering of learner activity in MOOCs.	Motivates treating low recorded activity as a behaviour pattern, not a fixed trait.
[9] Hlosta et al. (2017)	Early identification of at-risk students using legacy data without per-course models.	Predictive models on OULAD-style activity and assessment data.	Direct predictive baseline on the same kind of data; supports our checkpoint design.
[10] Kuzilek et al. (2017)	Describes the Open University Learning Analytics Dataset.	Dataset description and schema.	The dataset used in this project (Section 2).

2 Data, graphical model and fairness method

Data and estimand: Open University Learning Analytics Dataset [10], module BBB, loaded from a Kaggle copy (`anlgrbz/student-demographics-online-education-dataset`) . Pool: presentations 2013B, 2013J and 2014B, split 60/20/20 by student into train, validation and test. Transfer: presentation 2014J, with 81 students already seen in the pool removed. Population: students registered by day 14 and not yet unregistered. At checkpoint t, only students still registered at day $7t$ are eligible. Eligible students at week 2: 947 (validation), 945 (test), 1,723 (transfer). The outcome is adverse

for 46% of the test cohort. Protected attributes: gender, declared disability and age band. Primary groups: gender × disability (four cells). Secondary: age × disability; the 55-and-over band has no test students.

Graph and inference: The model is a DBN with education C (three levels) as root, a latent state Z_t with three states, weekly activity V_t (five categories: no activity, plus four bins of positive clicks) and an assessment-behind indicator Q_t (two levels, observed only in weeks with a due assessment). Y depends on the final state Z_T and on C. Inference follows the standard DBN treatment in Koller and Friedman [11] and Murphy [7]. Assumptions: Z_t is a statistical summary of recorded engagement, not ability or motivation; V_t and Q_t are conditionally independent given Z_t; no causal interpretation is claimed. Weeks after withdrawal are marginalised out.

$$p(C, Z_{1:T}, V_{1:T}, Q_{1:T}, Y) = \pi(Z_1 \mid C) \prod_{t=2}^{T} A(Z_t \mid Z_{t-1}) \prod_{t=1}^{T} B_V(V_t \mid Z_t)\, B_Q(Q_t \mid Z_t) \cdot \theta(Y \mid Z_T, C),$$

Query: $P(Y = 1 \mid C, V_{1:t}, Q_{1:t})$ for $t \in \{2, 4, 6, 8\}$, by forward filtering; EM for parameters.

Method and fairness: The proposed priority for student i at checkpoint j is $\text{logit}(p_{ij}) + \delta_{g_i,j} - \rho \cdot (\text{alerts already sent to } i)$, where δ holds group offsets for the four gender × disability cells at four checkpoints and ρ is a re-alert discount. Offsets and ρ are tuned on validation to minimise the cumulative coverage gap, checkpoint ΔEOpp and repeated-alert burden, with a penalty when detection falls more than 0.02 below the no-offset rule, and penalties on changes in δ across checkpoints. Primary criterion: cumulative coverage gap over supported groups, and checkpoint ΔEOpp. Diagnostics: ΔEOdds, ΔDP, PPV, false-alert exposure $P(\geq 1 \text{ alert} \mid Y = 0)$, time to first alert and repeated-alert fraction. Utility: cumulative detection $P(\geq 1 \text{ alert} \mid Y = 1)$ at fixed capacity, with acceptable loss 0.02. Group offsets need group membership at decision time, so the proposed rule is not group-blind. The DBN scores themselves are group-blind; a DBN with group input is a separate variant.

3 Evaluation protocol

Predictive baselines are logistic regression and gradient-boosted trees on checkpoint aggregates, and a static Bayesian network on cumulative observations. Probabilities are Platt-calibrated on validation. Fairness baselines are `indep_eopp` (independent per-checkpoint group offsets), `topk` (no offsets) and `single_threshold`. All policies are fitted on validation and replayed unchanged on test and transfer. Leakage control: checkpoint features use only days before $7t$, and features and eligibility at weeks 2, 4 and 6 are identical when all later events are deleted. Splits: five student-level split seeds (2024, 1, 2, 3, 4). Intervals are paired student-level bootstraps with 1,000 replicates, resampling students; the all-split comparison covers all five splits (Table 2), and the split-0 comparisons are reported separately. Holm correction is applied across the four confirmatory metrics (H1 coverage gap, H2 detection non-inferiority at margin 0.02, H3 time-to-first-alert gap, H4 false-alert exposure non-inferiority at margin 0.05); exploratory metrics are unadjusted. Model seeds: five DBN initialisation seeds were run on split 0 (Table 3). Tuning budgets were not matched: `indep_eopp` used 200 objective evaluations and `temporal` used 315, a deviation from the equal-budget requirement that is disclosed here. Exact inference: brute-force enumeration, three elimination orders and clique-tree message passing agree with the DBN's filter on 20 test students at four checkpoints, with tolerance 10^{-10} and maximum discrepancy 3.1×10^{-15}. Ablations (no transitions, no smoothness penalty, no latent state, no re-alert discount) and robustness checks (checkpoint schedules, two versus three states, 30% activity suppression, alternative outcomes, age × disability groups, capacities 10–30%) were run.

4 Results and interpretation

Method	Utility	Fairness gap	Log loss / Brier
Closest baseline (`indep_eopp`)	0.567 (0.017)	0.114 (0.096)	0.576 / 0.198
Proposed method (`temporal`)	0.683 (0.076)	0.095 (0.044)	0.576 / 0.198
Required ablation (`temporal_nosmooth`)	0.669 (0.101)	0.114 (0.054)	0.576 / 0.198

Table note: Test set, 20% capacity, mean (sd) over five student-level split seeds. Utility is cumulative detection $P(\geq 1 \text{ alert} \mid Y = 1)$, higher is better. Fairness gap is the max−min cumulative coverage over supported groups (at least 30 positives and negatives), lower is better. Log loss and Brier are for the DBN's calibrated probabilities, which all three policies share; Brier is the mean of the four weekly values on test. Split 0 alone: detection +0.194 (95% CI 0.161 to 0.231); coverage gap −0.173 (−0.328 to 0.028; Holm $p = 0.31$); false-alert exposure +0.273 (non-inferiority fails, Holm 1.00).

Table 2: Temporal rule minus baseline, mean over five student-level splits (negative is better for gaps and false-alert exposure). Significance uses the Holm-adjusted bootstrap p over H1–H4. For H2 and H4 the test is non-inferiority (margins 0.02 and 0.05), so "significant" means the margin holds.

Set	Baseline	Metric	Mean diff	SD of diff	Splits significant (of 5)
Test	`indep_eopp`	H1 coverage gap	−0.019	0.088	0
Test	`indep_eopp`	H2 detection	0.116	0.066	5
Test	`indep_eopp`	H3 time-to-first-alert gap	0.180	0.437	0
Test	`indep_eopp`	H4 false-alert exposure	0.134	0.093	1
Test	`topk`	H1 coverage gap	0.006	0.087	0
Test	`topk`	H2 detection	0.115	0.059	5
Test	`topk`	H3 time-to-first-alert gap	0.311	0.736	0
Test	`topk`	H4 false-alert exposure	0.136	0.095	1
Transfer	`indep_eopp`	H1 coverage gap	−0.114	0.164	1
Transfer	`indep_eopp`	H2 detection	0.166	0.122	5
Transfer	`indep_eopp`	H3 time-to-first-alert gap	−0.368	0.440	1
Transfer	`indep_eopp`	H4 false-alert exposure	0.178	0.134	1
Transfer	`topk`	H1 coverage gap	0.036	0.047	0
Transfer	`topk`	H2 detection	0.163	0.120	5
Transfer	`topk`	H3 time-to-first-alert gap	0.489	0.562	1
Transfer	`topk`	H4 false-alert exposure	0.177	0.136	1

Main finding: Relative to independent per-checkpoint equalisation, the temporal rule has higher detection in every split (test mean 0.683 vs 0.567; transfer mean 0.714 vs 0.548). Its mean coverage gap is lower (test 0.095 vs 0.114; transfer 0.138 vs 0.253), but the test difference is not significant on any split, and on transfer it is significant in one of five splits. On split 0 transfer, the gap is lower by 0.398 (Holm 0.006), and time to first alert is shorter by 0.93 weeks. Its false-alert exposure is much higher (test 0.300 vs 0.166; transfer 0.411 vs 0.233), so the non-inferiority margin of 0.05 fails on average. Against top-k, which ignores groups, the temporal rule does not reduce the gap (test 0.095 vs 0.088; transfer 0.138 vs 0.102), and its time to first alert is longer on transfer (+0.49 weeks on average; significant in one split). The fairness hypothesis is therefore not supported beyond the weaker baseline. At matched detection the advantage shrinks. Temporal at 20% (0.683) and independent at 30% (0.691) have similar gaps (0.222 vs 0.231) and exposure (0.300 vs 0.287) on test. Temporal at 10% reaches detection close to independent at 20% on transfer (0.567 vs 0.548) with a smaller gap (0.221 vs 0.267). These capacity-sweep means carry no intervals.

Mechanism and failure: The detection gain depends on the re-alert discount. Without it (`temporal_rho0`), detection is 0.570 on test, close to the baseline, and exposure falls from 0.300 to 0.170. Coverage-only

(discount without offsets) reaches detection 0.675 with exposure 0.291, so the discount drives most of the detection gain. The smoothness penalty lowers the checkpoint ΔEOpp (0.153 vs 0.200 without it) but slightly raises the transfer gap (0.138 vs 0.122). On split 0, the tuned policy uses only three non-zero offsets (all for the M|N cell at weeks 2 to 6, with values −0.50, −0.42 and −0.33), against seven or eight for the baseline. Across the five model seeds, the test temporal results are identical, the indep test policy uses 7 or 8 non-zero offsets depending on the seed, and transfer values differ at the third decimal (Table 3). The DBN's transitions help the likelihood (held-out weekly log-likelihood −1.852, against −2.130 with iid states and −1.872 with identity transitions), but the DBN is slightly below logistic regression on week-8 test AP (0.780 vs 0.794) and transfer AP (0.689 vs 0.691), and its week-8 log loss is higher than logistic regression's (0.529 vs 0.523 on test). The checkpoint-level equalisation failed to generalise: on transfer, split 0, the baseline's checkpoint ΔEOpp was 0.335, against 0.081 for top-k. Validation cells are very small (the validation M|Y cell has seven positives and one negative), so the offsets are poorly estimated. Alerts follow recorded activity and assessment status. On test, the DBN alerts 68% of bottom-quartile activity students at week 2, against 20% overall. At week 4, it alerts 94% of students with an assessment behind, against 11% of the others, and at week 8 it alerts no student without a missing assessment. Giving group membership to the DBN changes 10–30% of alerts depending on the week (Table 4). With group input, F|Y's week-2 alert rate rises from 0.289 to 0.434, and M|Y's week-8 rate from 0.308 to 0.538 (test). Subgroup sizes are small: test M|Y has 14 students (7 positives, 7 negatives), and transfer M|Y has 23, so these cells are exploratory.

Table 3: Five DBN initialisation seeds, split 0, 20% capacity. Each cell shows the minimum–maximum over the five seeds. Test temporal results are identical across seeds; small differences remain elsewhere.

Set	Policy	Detection	Coverage gap	False-alert exposure	ΔEOpp (mean)	Non-zero offsets
Test	`indep_eopp`	0.583	0.263	0.173	0.130	7–8
Test	`temporal`	0.778	0.090	0.446	0.087	3
Transfer	`indep_eopp`	0.546	0.468	0.238–0.239	0.334–0.335	7–8
Transfer	`temporal`	0.871	0.070	0.609	0.189–0.191	3

Table 4: Alerts with and without group membership given to the DBN, test set, 20% capacity. Share = changed alerts / alerts in either set.

Week	Alerts (A-blind)	Alerts (with A)	Changed alerts	Share changed
2	189	189	26	12.9%
4	186	186	44	21.2%
6	183	183	64	29.8%
8	179	179	18	9.6%

5 Limitations, ethics and conclusion

These results come from real OULAD records, but from a single module and a Kaggle copy whose version is not yet recorded. The five DBN seeds leave the test temporal results unchanged and move the other policies only slightly, so seed stability is supported on test but not yet fully verified across all policies. The alert simulator covers split 0 only. Several gender × disability cells are exploratory, and the 55-and-over band has no test students. Measurement is weak: recorded clicks may reflect offline study, assistive technology, work commitments or inaccessible materials, and missing assessments may reflect circumstances rather than disengagement. Repeated alerts may stigmatise students, so support must stay voluntary, with an opt-out and a route to contest an alert. The learned latent state is not causal. The conclusion is bounded: at 20% capacity, the temporal rule raises detection in every split and lowers the mean coverage gap relative to per-checkpoint equalisation, but the gap reduction is not reliable on test, false-alert exposure rises beyond the stated margin, and the rule does not beat top-k on fairness. The next experiment is to match tuning evaluations for both policies and to test the gap

with intervals across more splits and modules.

Reproducibility and individual contributions

Reproduce: `Runtime > Run all`, then cells A–I of `final(2).ipynb`; config: `cfg` dictionary in notebook Section 1. The notebook downloads the Kaggle data with `kagglehub`; set `OULAD_DIR` to use local CSVs. Environment: Google Colab;.
Anisha Chandra: Understanding and code implementation.
Dhanya Padasalgi: Research and documentation.
Evidence: `results/` (CSV and PDF outputs of cells A–I, including `paired_bootstrap_all_splits.csv`, `model_seeds.csv`, `logloss_brier.csv`, `group_input_alert_changes.csv`, `alert_policy_simulation_test.c` `temporal_fairness.pdf`).

Table 5: Alert-policy simulator, test set, split 0. Policies fitted on validation and replayed on test. Cumulative detection is $P(\geq 1 \text{ alert} \mid Y = 1)$.

Policy	Capacity	Week	Eligible	Alerted	Cum. detection	Cum. coverage gap	Checkpoint ΔEOpp
topk	10%	2	945	94	0.183	0.096	0.096
topk	10%	4	931	93	0.271	0.106	0.132
topk	10%	6	915	92	0.294	0.101	0.134
topk	10%	8	897	90	0.315	0.116	0.139
indep_eopp	10%	2	945	94	0.181	0.053	0.053
indep_eopp	10%	4	931	93	0.271	0.367	0.373
indep_eopp	10%	6	915	92	0.308	0.352	0.165
indep_eopp	10%	8	897	90	0.326	0.328	0.089
temporal	10%	2	945	94	0.183	0.080	0.080
temporal	10%	4	931	93	0.343	0.099	0.034
temporal	10%	6	915	92	0.500	0.113	0.014
temporal	10%	8	897	90	0.609	0.109	0.148
topk	20%	2	945	189	0.333	0.182	0.182
topk	20%	4	931	186	0.442	0.112	0.109
topk	20%	6	915	183	0.519	0.058	0.265
topk	20%	8	897	179	0.593	0.084	0.060
indep_eopp	20%	2	945	189	0.333	0.192	0.192
indep_eopp	20%	4	931	186	0.444	0.209	0.113
indep_eopp	20%	6	915	183	0.512	0.215	0.081
indep_eopp	20%	8	897	179	0.583	0.263	0.136
temporal	20%	2	945	189	0.333	0.192	0.192
temporal	20%	4	931	186	0.514	0.307	0.039
temporal	20%	6	915	183	0.662	0.108	0.107
temporal	20%	8	897	179	0.778	0.090	0.011
topk	30%	2	945	284	0.470	0.167	0.167
topk	30%	4	931	279	0.562	0.144	0.096
topk	30%	6	915	274	0.613	0.138	0.102
topk	30%	8	897	269	0.694	0.181	0.029
indep_eopp	30%	2	945	284	0.468	0.301	0.301
indep_eopp	30%	4	931	279	0.553	0.337	0.210
indep_eopp	30%	6	915	274	0.618	0.156	0.102
indep_eopp	30%	8	897	269	0.694	0.218	0.211
temporal	30%	2	945	284	0.470	0.167	0.167
temporal	30%	4	931	279	0.713	0.102	0.093
temporal	30%	6	915	274	0.843	0.023	0.216
temporal	30%	8	897	269	0.900	0.036	0.116

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